

Siddhant K. Sancheti

Software Engineer | Full Stack, AI/ML & Distributed Systems
+1 (669) 319-8812 | siddhantsanchetik@gmail.com | [LinkedIn](#) | [Portfolio](#)

SUMMARY

Software engineer with 3+ years across backend, infrastructure, full-stack, and AI/ML - building distributed systems in Java and Python at AWS, and developing AI/ML systems end-to-end, from training computer vision models to engineering production GenAI pipelines. Cut engineering delivery cycles ~50x with a Python/Go multi-agent orchestration system and led a zero-downtime IAM migration across 110+ Lambda services in 17 regions. Shipped full-stack products (React, Node) and GPT-4/LangChain RAG pipelines that cut latency 4x and lifted product outcomes 85% across systems serving 5,000+ hosts and 100,000+ users.

WORK EXPERIENCE

Amazon Web Services Seattle, WA
Software Development Engineer May 2025 - Present

- Co-designed a Python-based agentic workflow system of 5-10 parallel agents with persistent session state and resumable, asynchronous execution, converting task definitions into dependency DAGs and reducing 10-task delivery cycles across 195+ packages from 5-7 days to 1-2 hours.
- Migrated password-based authentication across 110+ AWS Lambda services in 17 regions to AWS IAM with zero downtime and zero password fallback, sustaining 150-190 database connections per second and achieving 100% IAM adoption.
- Built a self-service fleet capacity management platform using Java, Amazon DynamoDB, RPC APIs, and production-safe CLI tools, reducing scaling workflows from 30 minutes to 2-4 minutes across 5,000+ EC2 hosts.
- Strengthened service reliability by resolving 110+ production incidents through on-call rotations, triage, root cause analysis, and controlled release management, cutting recurring failures through rollback planning and customer coordination.

Mercor San Jose, CA
Web Development and Design (Contract) Jan. 2025 - May 2025

- Evaluated 100+ AI-generated React applications across 20+ criteria, including functionality, performance, accessibility, responsiveness, and UI/UX quality, identifying defects and edge cases to ensure alignment with user requirements.
- Reviewed design consistency and accessibility standards across 50+ features, identifying opportunities for improvement and maintaining high-quality user experiences.
- Provided structured feedback loops to project teams across 15+ sprints, continuously refining features and design elements to meet evolving user needs and industry best practices.

GrantAide San Jose, CA
Software Engineer - ML/AI (Contract) Sep. 2024 - Jan. 2025

- Built and scaled an AI-powered grant-writing platform using Python, Flask, React, GPT-4, Vertex AI, LangChain, FAISS, AWS (Elastic Beanstalk, Amplify, S3, EC2), and Stripe, improving platform responsiveness by 5x and supporting subscription and payment workflows.
- Engineered retrieval-augmented generation pipelines that reduced query latency from 200 milliseconds to 50 milliseconds, increased response accuracy by 80%, and shortened frontend load times by 35%.
- Developed the frontend using React and Material UI with Firebase for authentication and real-time data sync, improving component reusability and delivering a responsive, accessible user experience across the platform.

Forsk Technologies Jaipur, India
Full-Stack Software Developer May 2021 - Jul. 2022

- Architected an AI-enabled campus platform using React, Node.js, AWS, and a TensorFlow/PyTorch-based chatbot and feedback system for student support, reducing onboarding time by 50% for 5,000+ monthly active users and increasing service adoption by 35%.
- Redesigned GraphQL-based data access and application workflows, reducing data loading time by 20% and improving frontend performance by 30%.
- Established CI/CD pipelines, containerized deployments, and production monitoring using Jenkins, GitHub Actions, Docker, and Grafana, increasing release frequency by 40% while maintaining 99.9% uptime.

Pantech Prolabs Pune, India
Software Engineer Aug. 2020 - Feb. 2021

- Designed asynchronous ETL data pipelines using Python, Apache Kafka, Apache ZooKeeper, and Apache Spark, reducing processing time by 30% and supporting large-scale job listing data ingestion and transformation.
- Secured REST APIs using Django, Flask, OAuth 2.0, and JWT for 100,000+ users, reducing unauthorized access incidents by 45% and cutting infrastructure provisioning effort by 35% through Terraform automation.

PROJECTS

Agentic AI Career Operations Platform

- Co-developed an agentic AI job search platform using Go, JavaScript, Claude Code, Playwright, and GitHub Actions to automate job discovery, resume tailoring, and application tracking across 500+ companies, enabling parallel job evaluation and reducing manual effort by 80%.

Advanced Lane Detection for Autonomous Vehicles

- Engineered a computer vision lane-detection system using CARLA, ENet, Hourglass neural networks, and RANSAC, achieving 93.19% classification accuracy and 3.89% mean absolute error across varied driving conditions.

SKILLS

Programming Languages	: Python, Java, Go, Ruby, TypeScript, JavaScript, SQL, Bash, Shell Scripting, HTML5, CSS3
Backend & Web Frameworks	: Spring Boot, Node.js, Express.js, Flask, Django, React, Bootstrap, REST APIs, Remote Procedure Call (RPC), GraphQL
Cloud & Infrastructure	: Amazon Web Services (AWS), Google Cloud Platform (GCP), AWS Lambda, Amazon EC2, Amazon S3, Amazon ECS, AWS Bedrock, Amazon VPC, Amazon EBS, Amazon Route 53, AWS Cloud Development Kit (CDK), Kubernetes, Docker, Terraform, Linux
Data Engineering & Messaging	: Apache Spark, Apache Kafka, Apache ZooKeeper, Asynchronous Programming, Concurrent Programming, Data Pipelines, ETL
Databases	: PostgreSQL, Amazon DynamoDB, Amazon RDS, MongoDB, Firebase
Artificial Intelligence & Machine Learning	: TensorFlow, PyTorch, JAX, Keras, TensorRT, OpenCV, Computer Vision, Natural Language Processing (NLP), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangChain, Model Context Protocol (MCP), Vertex AI, Amazon SageMaker, FAISS, Pandas, NumPy
DevOps & Observability	: CI/CD, Jenkins, GitHub Actions, Git, Amazon CloudWatch, AWS CloudTrail, Grafana, LangSmith, Production Monitoring, Logging, Incident Management
Tools & Methodologies	: Jira, Agile, Root Cause Analysis, Production Support, Release Management

EDUCATION

San José State University, San Jose, CA Aug. 2022 - May 2024
Master of Science in Software Engineering

Savitribai Phule Pune University, Pune, India Aug. 2016 - May 2020
Bachelor of Engineering, Electronics and Telecommunications